Text 1 Icebergs

Icebergs are among nature's most spectacular creations, and yet most people have never seen one. A vague air of mystery envelops them. They come into being somewhere-in faraway, frigid waters, amid thunderous noise and splashing turbulence, which in most cases no one hears or sees. They exist only a short time and then slowly waste away just unnoticed.

Objects of sheerest beauty, they have been called. Appearing in an endless variety of shapes they may be dazzlingly white, or they may be glassy blue, green or purple, tinted faintly or in darker hues. They are graceful, stately, inspiring-in calm, sunlit seas.

But they are also called frightening in that they can be found in the night, in the fog, and in storms. Even in clear weather one is wise to stay a safe distance away from them. Most of their bulk is hidden below the water, so their underwater parts may extend out far beyond the visible top. Also, they may roll over unexpectedly, churning the waters around them.

Icebergs are parts of glaciers that break off, drift into the water, float about awhile, and finally melt. Icebergs afloat today are made of snowflakes that have fallen over long ages of time. They embody snows that drifted down hundreds, or many thousands, or in some cases maybe a million years ago. The snows fell in Polar Regions and on cold mountains, where they melted only a little or not at all, and so collected to great depths over the years and centuries.

As each year's snow accumulation lay on the surface, evaporation and melting caused the snowflakes slowly to lose their feathery points and become tiny grains of ice. When new snow fell on top of the old, it too turned to icy grains. So blankets of snow and ice grains mounted layer upon layer and were of such great thickness that the weight of the upper layers compressed the lower ones. With time and pressure from above, the many small ice grains joined and changed to larger crystals, and eventually the deeper crystals merged into a solid mass of ice.

Text 2 The Effects of Pressure on the Bodies of Divers

Under certain circumstances, the human body must cope with gases at greater-than normal atmospheric pressure. For example, gas pressures increase rapidly during a dive made with scuba gear because the breathing equipment allows divers to stay underwater longer and dive deeper. The pressure exerted on the human body increases by 1 atmosphere for every 10 meters of depth in seawater, so that at 30 meters in seawater a diver is exposed to a pressure of about 4 atmospheres. The pressure of the gases being breathed must equal the external pressure applied to the body; otherwise breathing is very difficult. Therefore all of the gases in the air breathed by a scuba diver at 40 meters are present at five times their usual pressure. Nitrogen, which composes 80 percent of the air we breathe, usually causes a balmy feeling of well-being at this pressure. At a depth of 5 atmospheres, nitrogen causes symptoms resembling alcohol intoxication, known as nitrogen narcosis. Nitrogen narcosis apparently results from a direct effect on the brain of the large amounts of nitrogen dissolved in the blood. Deep dives are less dangerous if helium is substituted for nitrogen, because under these pressures helium does not exert a similar narcotic effect.

As a scuba diver descends, the pressure of nitrogen in the lungs increases. Nitrogen then diffuses from the lungs to the blood, and from the blood to body tissues. The reverse occurs when the diver surfaces; the nitrogen pressure in the lungs falls and the nitrogen diffuses from the tissues into the blood, and from the blood into the lungs. If the return to the surface is too rapid, nitrogen in the tissues and blood cannot diffuse out rapidly enough and nitrogen bubbles are formed. They can cause severe pains, particularly around the joints.

Another complication may result if the breath is held during ascent. During ascent from a depth of 10 meters, the volume of air in the lungs will double because the air pressure at the surface is only half of what it was at 10 meters. This change in volume may cause the lungs to distend and even rupture. This condition is called air embolism. To avoid this event, a diver must ascend slowly, never at a rate exceeding the rise of the exhaled air bubbles, and must exhale during ascent.